

To produce electronic assemblies without a known profile is just asking for trouble

Mark Stansfield, director, SolderStar speaks to Abhishek Mutha from EFY about the importance of controlled thermal processes and lead-free manufacturing and throws light on necessity of temperature profiling for electronic assemblies, different kinds of profiling and their benefits, and the future of thermal profiling

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Q. Why is thermal profiling necessary for lead-free manufacture?

When lead-free emerged as an established process, many manufacturers had major upheaval of their well-proven thermal processes. The available processing window is much tighter with lead-free, making the 'one profile fits all' much less likely. Profiling tools are now extremely advanced such that we can take a snapshot of what the thermal process is doing today and, using software simulation tools, it is possible to produce the optimal settings for engineering in a matter of minutes. Without correct settings on the machines, solder quality is impaired and components could be damaged.

Q. What is the difference between the two main kinds of profiles used today: Ramp-Soak-Spike (RSS) and Ramp-to-Spike (RTS)?

Use of an RSS or RTS profile is specific to the assembly being processed by the manufacturer. RTS is very common because it is quite simple to set up and easily achievable on most reflow ovens, however it is only suitable when the assembly does not have high mass elements within its design. If the assembly has big mass variations, or has high mass elements like aluminium PCB or large arrays, it may be necessary to employ RSS style profile. This technique has a more aggressive ramp, but longer soak period to allow the assembly temperature to more readily equalise before moving into the Spike/Reflow stage. Without this method, peak and time above liquidous temperatures within specification will not be possible due to large temperature deltas across the PCB.

Q. How does software complement thermal profiling equipment?

Profiling systems are definitely 50% a software product, if not more. Tools automatically calculate all process parameters from the raw data and display them against the working process windows defined by the solder paste or component specification, making the good/bad analysis very quick and easy. We have optimisation software that helps with finding optimal process settings. Additionally, we have a complete suite of QA tools to allow full documen-

tation of the profiling process and allow generation of SPC information from the data captured on the production lines.

Q. What is the significance of temperature-profiling systems?

I am a great believer in 'you can't manage what you don't measure.' To produce electronic assemblies without a known profile is just asking for trouble further down the line—either rework costs or early failures and unhappy customers. Not every company warrants investment in equipment to continuously monitor the production lines, but to not profile an assembly during the NPI stage and then performing some form of periodic checking is, to be frank, just not doing the job correctly.

Q. Why taking temperature profile measurements as the benchmark for process control not enough?

A soldering process is a dynamic thing, with many outside influences affecting the profile seen at product level, machine loading, flux build up, extraction performance change or machine failure. All these things can be measured and controlled with the adoption of a quality control procedure. As a minimum, verifying the machine before a new batch is sent through the line is common sense, but for long running production batches periodic checking would also be prudent.

Q. What is the future of thermal profiling?

The procedure of capturing a thermal profile, performing an optimisation step and verification of that step is well established and unlikely to change radically. What we are seeing from our customer base, especially with the automotive industries, is the need to measure and improve every part of the process, from performing initial machine capability to studies, to production level daily testing of all parameters within the manufacturing process. This need is central to our drive to provide measurement solutions within a common software platform for all the soldering techniques, additionally to provide the engineer with flexible instrumentation they can tailor exactly to their needs, both quickly and cost effectively. ●